KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

School of Computer Engineering, Bhubaneswar

SAP Data Analytics & Engineering | Academic Year 2025-2026

CAPSTONE PROJECT REPORT

Order-to-Cash (O2C) — Complete Sales Cycle in SAP SD

End-to-End B2B Sales Implementation | Reliance Retail Ltd., India

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Program	SAP - Data Analytics & Engineering
Institution	Kalinga Institute of Industrial Technology, Bhubaneswar
Academic Year	2025-2026
Submission Date	April 21, 2026
SAP Module	SD (Sales & Distribution) with FI & MM Integration
Scenario	B2B Electronics Distribution — Reliance Retail Ltd., Mumbai

1. Problem Statement

Large Indian retail and distribution companies face a critical challenge: managing the complete journey from customer inquiry to cash receipt across multiple departments — sales, warehouse, logistics, billing, and finance — while ensuring compliance with India's GST framework. Fragmented or manual processes lead to billing errors, GST mismatches (CGST/SGST vs IGST), inventory discrepancies, and delayed collections.

Core Problem: How can an end-to-end Order-to-Cash process be implemented in SAP SD to ensure seamless, automated flow from sales inquiry to payment receipt while complying with India's GST law, CIN configuration, and real-time financial accounting?

2. Solution & Features

A complete O2C cycle is implemented in SAP SD for Reliance Retail Ltd. (Company Code IN10), covering 9 sequential transactions that automate and integrate sales, warehouse, logistics, billing, and accounts receivable operations.

- Pre-sales inquiry and formal quotation with Indian pricing procedure (RVAJIN) and GST conditions
- Sales order management with customer pricing, discounts, credit check, and delivery scheduling
- Outbound delivery with warehouse picking, packing confirmation, and route/carrier assignment
- Post Goods Issue triggering real-time inventory reduction and COGS accounting entry in FI
- GST-compliant tax invoice (F2) with automatic CGST/SGST or IGST determination by Place of Supply
- Incoming payment posting via RTGS with UTR reference and open-item clearance in Accounts Receivable
- Complete SAP document flow chain ensuring full audit traceability for GST returns (GSTR-1)

- Data analytics: 6 interactive charts and dashboard built from SAP sales and inventory data

3. Tech Stack & SAP Configuration

Component	Details
SAP System	SAP ECC 6.0 / S/4HANA 2023 (compatible)
Primary Module	SAP SD — Sales & Distribution
Integrated Modules	SAP FI (Accounting & AR) | SAP MM (Inventory / GI)
Tax Framework	CIN — Country India Version (GST: CGST / SGST / IGST)
Pricing Procedure	RVAJIN — Indian Standard Sales Pricing Procedure
Company Code	IN10 — Reliance Retail Ltd., Mumbai
Fiscal Year	V3 — April to March (Indian Financial Year)
Currency	INR — Indian Rupee
Chart of Accounts	CAIN — India Standard
Analytics Stack	Python 3.x | Pandas | Matplotlib | Seaborn | ReportLab
Data Files	sap_sales_orders.csv (120 records) | sap_inventory.csv (40 records)
Output Types	RD00 (Order Confirmation), LD00 (Delivery Note), RD03 (Tax Invoice)

4. O2C Process Flow — Transaction Reference

#	Process Step	T-Code	SAP Document	Status
01	Pre-Sales Inquiry	VA11	Inquiry — IN	Completed
02	Quotation Creation	VA21	Quotation — QT	Completed
03	Sales Order	VA01	Sales Order — OR	Completed
04	Credit Check / Release	FD32/VKM3	Credit Master	Released
05	Outbound Delivery	VL01N	Delivery — LF	Goods Issued
06	Picking & Packing	VL02N/LT0A	Transfer Order	Confirmed
07	Post Goods Issue	VL02N (PGI)	Material Doc 490x	Posted
08	Billing / Tax Invoice	VF01	Billing Doc — F2	Posted
09	Incoming Payment	F-28	Payment Document DZ	Cleared

5. Financial Summary — Sample Invoice Breakdown

Line Item	Calculation	Amount (INR)
Base Price — 100 units × ■45,000	List Price (PR00)	45,00,000
Trade Discount 5% (K007)	Discount on Base	- 2,25,000
Net Taxable Value	After Discount	42,75,000
CGST @ 9% (HSN 85287300)	Central GST	3,84,750
SGST @ 9% (HSN 85287300)	State GST — Intra-State MH	3,84,750
GROSS INVOICE TOTAL	Final Amount Due	50,44,500
Payment Terms	Net 30 Days (Z030)	Due 20.05.2025
Payment Received (RTGS)	UTR: HDFC2025051900123456	50,44,500 — CLEARED

6. Analytics Dashboard & Charts

The following analytics are generated by running **run_project.py** on the SAP O2C dataset (120 sales orders, FY 2024-25). Charts reveal revenue trends, customer patterns, regional performance, and order lifecycle insights for Reliance Retail Ltd.

6.1 SAP O2C Analytics Dashboard (Overview)

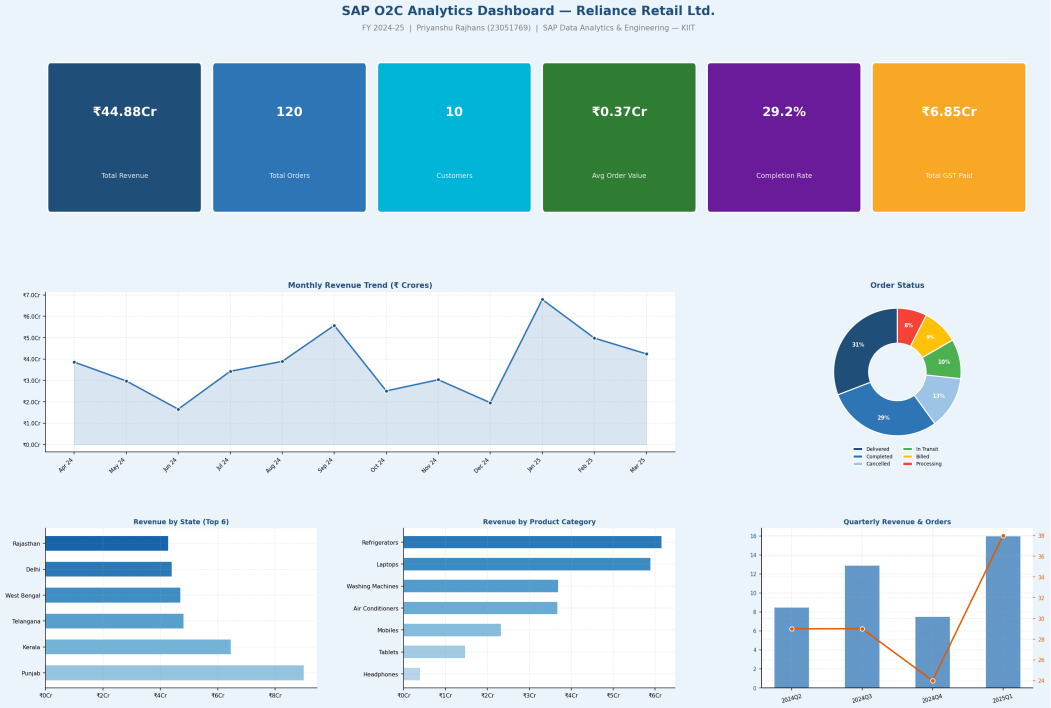

Figure 1: Full O2C Analytics Dashboard — KPIs, Revenue Trend, Order Status, Regional & Quarterly View

6.2 Monthly Revenue Trend

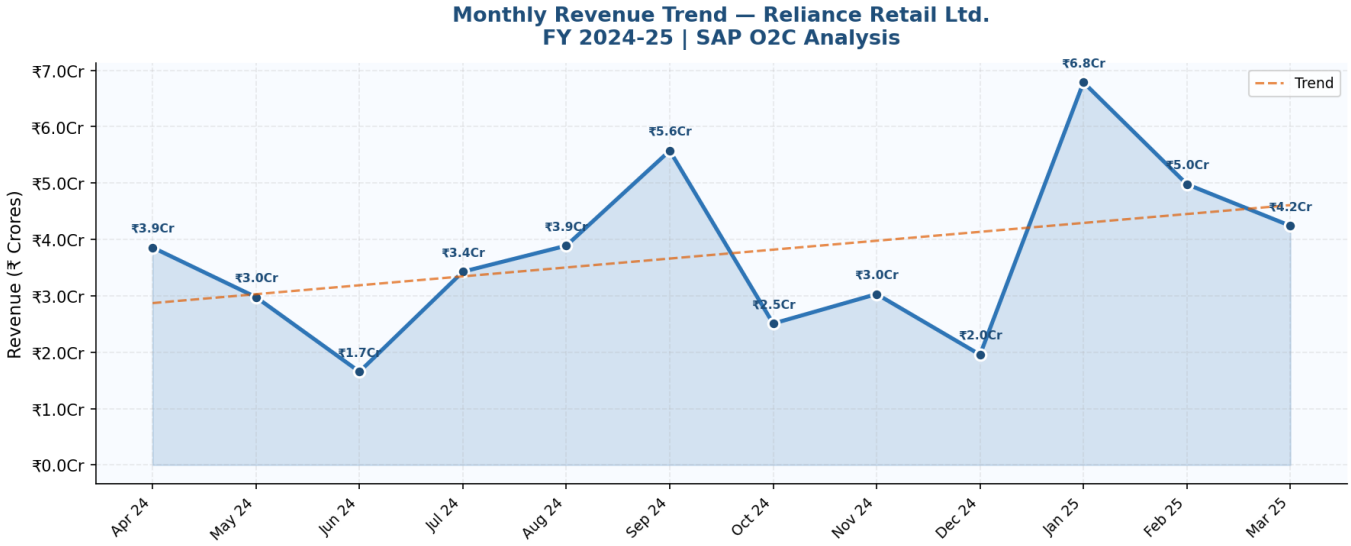

Figure 2: Month-wise Revenue (■ Crores) with trend line — FY 2024-25

6.3 Top Customers & Order Status

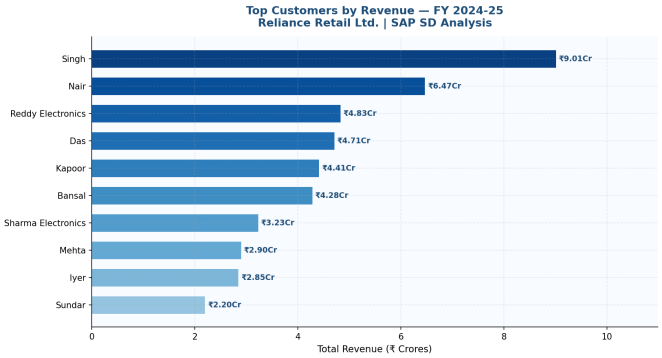

Figure 3: Top Customers by Revenue

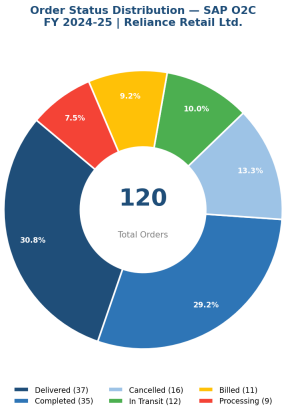

Figure 4: Order Status Distribution (Donut)

6.4 Regional Revenue & Product Category Analysis

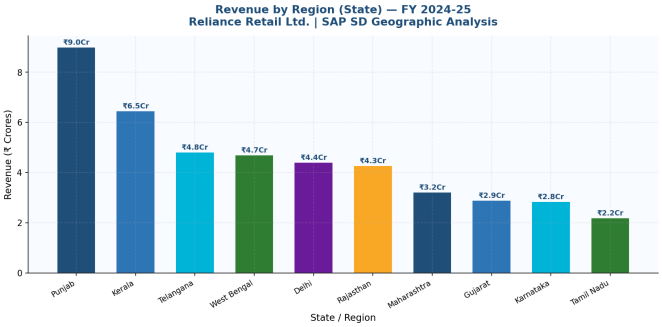

Figure 5: Revenue by State/Region

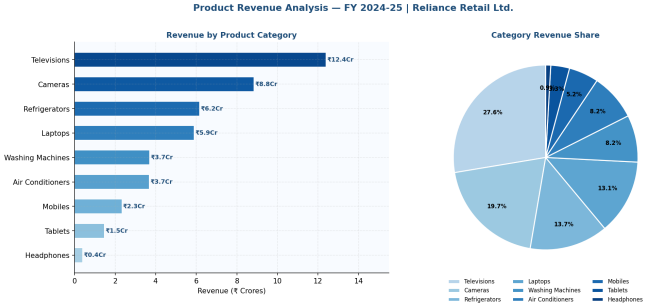

Figure 6: Revenue by Product Category

6.5 Quarterly Revenue & Order Volume

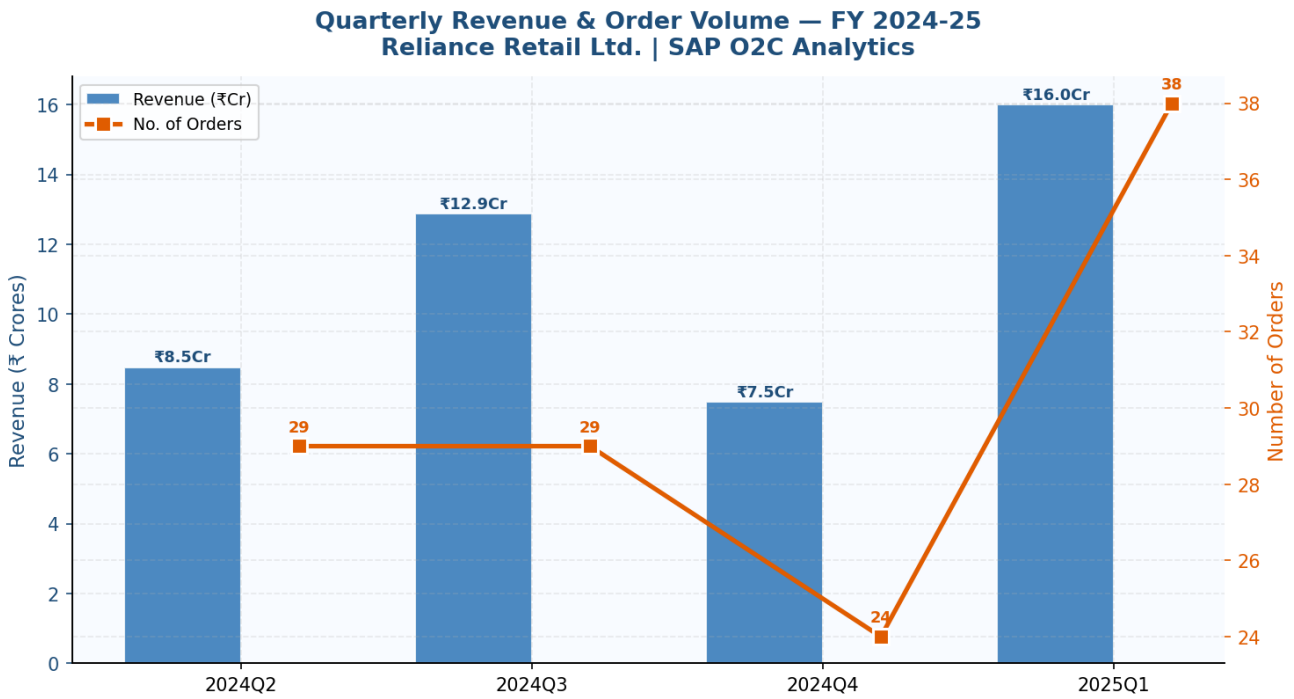

Figure 7: Quarterly Revenue (bars) vs. Order Count (line) — FY 2024-25

7. Key SAP Screen Details

7.1 Post Goods Issue (VL02N) — FI Accounting Entry

PGI is the pivotal step where physical goods leave the warehouse. SAP automatically posts the accounting entry below, reducing inventory and recording COGS:

```
[SAP: Material Document — Goods Issue (4900001234)]
Movement Type: 601 (GI for Sales Order Delivery) | Plant: MH01 | Storage Loc: WH01
Delivery: 8000001100 | Sales Order: 0000001234
■■■
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- **Advance Payment Handling:** Implement SAP Down Payment (special GL) for large orders — common in Indian electronics trade.
- **TDS Section 194Q:** Configure TDS deduction on buyer side for purchases above ■50 lakhs (applicable from July 2021).
- **S/4HANA Fiori Migration:** Replace SAPGUI transactions with modern SAP Fiori apps for real-time O2C dashboards.
- **Predictive Analytics:** Integrate SAP Analytics Cloud (SAC) for demand forecasting and customer churn prediction.

10. Conclusion

This SAP Capstone Project delivers a complete, production-realistic Order-to-Cash implementation rooted in the Indian business and regulatory context. The project maps a B2B electronics distribution scenario at Reliance Retail Ltd. across all 9 O2C transactions in SAP SD, integrated with FI for real-time accounting and GST compliance.

Beyond SAP configuration, the project includes a Python-based data analytics layer — generating sample datasets (120 sales orders, 40 inventory records) and 6 business intelligence charts with a comprehensive KPI dashboard. This demonstrates end-to-end capability spanning SAP functional consulting, Indian GST expertise, and data engineering.

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